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Instruction Format Specification – IA-32

1. Introduction

This document presents a concise specification of selected IA-32 instruction encodings for a two-pass assembler. It describes how assembly instructions are represented as variable-length machine-code sequences and identifies the fields that may be required during encoding, including the opcode, ModR/M byte, SIB byte, immediate values, and displacements. The reference table provides the operand forms and encoding information needed to analyse and construct the corresponding 32-bit x86 instruction bytes.

2. IA-32 Instruction Byte Diagram

An IA-32 instruction is variable length. Depending on the instruction and addressing mode, optional fields may follow the opcode.

Opcode	ModR/M	SIB	Displacement	Immediate
Mandatory	Optional	Optional	Optional	Optional

Instruction layout: Opcode → ModR/M → SIB → Displacement → Immediate

3. Encoding Field Meaning

Field	Meaning
ModR/M	Contains MOD, REG and R/M fields and identifies register or memory operands.
SIB	Scale-Index-Base byte used for complex memory addressing.
+rd	Used by opcode families such as B8+rd, where the register number is added to the base opcode.
ib	8-bit immediate.
iw	16-bit immediate.
id	32-bit immediate.
cb	8-bit relative displacement.

cw	16-bit relative displacement.
cd	32-bit relative displacement.

4. Common IA-32 Opcode / Encoding Table

Mnemonic	Operand 1	Operand 2	ModR/M	+rd	SIB	Immediate	Opcode
MOV	r/m8	r8	/r	0	Yes	—	88 /r
MOV	r/m16	r16	/r	0	Yes	—	89 /r
MOV	r/m32	r32	/r	0	Yes	—	89 /r
MOV	r8	r/m8	/r	0	Yes	—	8A /r
MOV	r16	r/m16	/r	0	Yes	—	8B /r
MOV	r32	r/m32	/r	0	Yes	—	8B /r
MOV	r/m8	imm8	/0	0	Yes	ib	C6 /0 ib
MOV	r/m16	imm16	/0	0	Yes	iw	C7 /0 iw
MOV	r/m32	imm32	/0	0	Yes	id	C7 /0 id
MOV	r8	imm8	0	0	No	ib	B0+rb ib
MOV	r16	imm16	0	0	No	iw	B8+rw iw
MOV	r32	imm32	0	1	No	id	B8+rd id
ADD	r/m8	r8	/r	0	Yes	—	00 /r
ADD	r/m16	r16	/r	0	Yes	—	01 /r
ADD	r/m32	r32	/r	0	Yes	—	01 /r
ADD	r8	r/m8	/r	0	Yes	—	02 /r
ADD	r16	r/m16	/r	0	Yes	—	03 /r
ADD	r32	r/m32	/r	0	Yes	—	03 /r
ADD	r/m8	imm8	/0	0	Yes	ib	80 /0 ib
ADD	r/m16	imm16	/0	0	Yes	iw	81 /0 iw
ADD	r/m32	imm32	/0	0	Yes	id	81 /0 id
ADD	r/m16	imm8	/0	0	Yes	ib	83 /0 ib
ADD	r/m32	imm8	/0	0	Yes	ib	83 /0 ib
ADD	AL	imm8	0	0	No	ib	04 ib
ADD	AX	imm16	0	0	No	iw	05 iw
ADD	EAX	imm32	0	0	No	id	05 id
SUB	r/m8	r8	/r	0	Yes	—	28 /r
SUB	r/m16	r16	/r	0	Yes	—	29 /r
SUB	r/m32	r32	/r	0	Yes	—	29 /r
SUB	r8	r/m8	/r	0	Yes	—	2A /r
SUB	r16	r/m16	/r	0	Yes	—	2B /r
SUB	r32	r/m32	/r	0	Yes	—	2B /r
SUB	r/m8	imm8	/5	0	Yes	ib	80 /5 ib
SUB	r/m16	imm16	/5	0	Yes	iw	81 /5 iw
SUB	r/m32	imm32	/5	0	Yes	id	81 /5 id
SUB	r/m16	imm8	/5	0	Yes	ib	83 /5 ib
SUB	r/m32	imm8	/5	0	Yes	ib	83 /5 ib
AND	r/m8	r8	/r	0	Yes	—	20 /r
AND	r/m16	r16	/r	0	Yes	—	21 /r
AND	r/m32	r32	/r	0	Yes	—	21 /r
AND	r8	r/m8	/r	0	Yes	—	22 /r
AND	r16	r/m16	/r	0	Yes	—	23 /r
AND	r32	r/m32	/r	0	Yes	—	23 /r
AND	r/m8	imm8	/4	0	Yes	ib	80 /4 ib
AND	r/m16	imm16	/4	0	Yes	iw	81 /4 iw
AND	r/m32	imm32	/4	0	Yes	id	81 /4 id
AND	r/m16	imm8	/4	0	Yes	ib	83 /4 ib
AND	r/m32	imm8	/4	0	Yes	ib	83 /4 ib
OR	r/m8	r8	/r	0	Yes	—	08 /r

OR	r/m16	r16	/r	0	Yes	—	09 /r
OR	r/m32	r32	/r	0	Yes	—	09 /r
OR	r8	r/m8	/r	0	Yes	—	0A /r
OR	r16	r/m16	/r	0	Yes	—	0B /r
OR	r32	r/m32	/r	0	Yes	—	0B /r
OR	r/m8	imm8	/1	0	Yes	ib	80 /1 ib
OR	r/m16	imm16	/1	0	Yes	iw	81 /1 iw
OR	r/m32	imm32	/1	0	Yes	id	81 /1 id
OR	r/m16	imm8	/1	0	Yes	ib	83 /1 ib
OR	r/m32	imm8	/1	0	Yes	ib	83 /1 ib
XOR	r/m8	r8	/r	0	Yes	—	30 /r
XOR	r/m16	r16	/r	0	Yes	—	31 /r
XOR	r/m32	r32	/r	0	Yes	—	31 /r
XOR	r8	r/m8	/r	0	Yes	—	32 /r
XOR	r16	r/m16	/r	0	Yes	—	33 /r
XOR	r32	r/m32	/r	0	Yes	—	33 /r
XOR	r/m8	imm8	/6	0	Yes	ib	80 /6 ib
XOR	r/m16	imm16	/6	0	Yes	iw	81 /6 iw
XOR	r/m32	imm32	/6	0	Yes	id	81 /6 id
XOR	r/m16	imm8	/6	0	Yes	ib	83 /6 ib
XOR	r/m32	imm8	/6	0	Yes	ib	83 /6 ib
CMP	r/m8	r8	/r	0	Yes	—	38 /r
CMP	r/m16	r16	/r	0	Yes	—	39 /r
CMP	r/m32	r32	/r	0	Yes	—	39 /r
CMP	r8	r/m8	/r	0	Yes	—	3A /r
CMP	r16	r/m16	/r	0	Yes	—	3B /r
CMP	r32	r/m32	/r	0	Yes	—	3B /r
CMP	r/m8	imm8	/7	0	Yes	ib	80 /7 ib
CMP	r/m16	imm16	/7	0	Yes	iw	81 /7 iw
CMP	r/m32	imm32	/7	0	Yes	id	81 /7 id
CMP	r/m16	imm8	/7	0	Yes	ib	83 /7 ib
CMP	r/m32	imm8	/7	0	Yes	ib	83 /7 ib
NOT	r/m8	—	/2	0	Yes	—	F6 /2
NOT	r/m16	—	/2	0	Yes	—	F7 /2
NOT	r/m32	—	/2	0	Yes	—	F7 /2
MUL	r/m8	—	/4	0	Yes	—	F6 /4
MUL	r/m16	—	/4	0	Yes	—	F7 /4
MUL	r/m32	—	/4	0	Yes	—	F7 /4
IMUL	r/m8	—	/5	0	Yes	—	F6 /5
IMUL	r/m16	—	/5	0	Yes	—	F7 /5
IMUL	r/m32	—	/5	0	Yes	—	F7 /5
IMUL	r16	r/m16	/r	0	Yes	—	0F AF /r
IMUL	r32	r/m32	/r	0	Yes	—	0F AF /r
IMUL	r16	r/m16	/r	0	Yes	ib	6B /r ib
IMUL	r32	r/m32	/r	0	Yes	ib	6B /r ib
IMUL	r16	r/m16	/r	0	Yes	iw	69 /r iw
IMUL	r32	r/m32	/r	0	Yes	id	69 /r id
DIV	r/m8	—	/6	0	Yes	—	F6 /6
DIV	r/m16	—	/6	0	Yes	—	F7 /6
DIV	r/m32	—	/6	0	Yes	—	F7 /6
INC	r/m8	—	/0	0	Yes	—	FE /0
INC	r/m16	—	/0	0	Yes	—	FF /0

INC	r/m32	—	/0	0	Yes	—	FF /0
INC	r16	—	0	0	No	—	40+rw
INC	r32	—	0	1	No	—	40+rd
DEC	r/m8	—	/1	0	Yes	—	FE /1
DEC	r/m16	—	/1	0	Yes	—	FF /1
DEC	r/m32	—	/1	0	Yes	—	FF /1
DEC	r16	—	0	0	No	—	48+rw
DEC	r32	—	0	1	No	—	48+rd
JMP	rel8	—	0	0	No	cb	EB cb
JMP	rel16	—	0	0	No	cw	E9 cw
JMP	rel32	—	0	0	No	cd	E9 cd
JMP	r/m16	—	/4	0	Yes	—	FF /4
JMP	r/m32	—	/4	0	Yes	—	FF /4
JZ	rel8	—	0	0	No	cb	74 cb
JZ	rel16/32	—	0	0	No	cw/cd	0F 84 cw/cd
JNZ	rel8	—	0	0	No	cb	75 cb
JNZ	rel16/32	—	0	0	No	cw/cd	0F 85 cw/cd
CALL	rel16	—	0	0	No	cw	E8 cw
CALL	rel32	—	0	0	No	cd	E8 cd
CALL	r/m16	—	/2	0	Yes	—	FF /2
CALL	r/m32	—	/2	0	Yes	—	FF /2
RET	—	—	0	0	No	—	C3
RET	imm16	—	0	0	No	iw	C2 iw